Physio-BP: Physiologically-Guided Dual Radar System for Continuous Blood Pressure Estimation

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***Abstract*—Continuous blood pressure (BP) estimation remains a significant challenge in non-contact hemodynamic monitoring, where existing methods struggle to strike a balance between clinical accuracy and usability.We present Physio-BP, a physiologically-guided dual-radar system enabling calibration-free, continuous tracking of systolic (SBP) and diastolic blood pressure (DBP) by integrating spatially resolved vascular dynamics. Unlike conventional single-waveform radar approaches, Physio-BP synchronously captures mechanical signals from the heart and pulse waves from the carotid artery, allowing precise measurement of pulse arrival time (PAT) and inter-beat intervals (IBI). To enhance signal quality, we introduce a novel joint signal selection algorithm that optimizes feature extraction across dual radars, effectively addressing the inherent signal-to-noise ratio (SNR) disparities between the chest and neck monitoring sites. To validate our system, an extensive dataset exceeding 50 hours was collected under diverse conditions, including different seasons, times of day, body postures, and radar devices, ensuring comprehensive coverage. Experimental results on the dataset confirm that Physio-BP enables continuous and precise SBP and DBP tracking, a capability not achieved by current baseline methods, thereby advancing non-contact sensing for hemodynamic monitoring.**

***Index Terms*—Continuous blood pressure monitoring, dual-radar system, non-contact sensing, pulse transit time, systolic and diastolic blood pressure.**

I. Introduction

THE shift from intermittent to continuous hemodynamic monitoring is crucial for early detection of cardiovascular risks and offers clear benefits, including improved diagnostic accuracy, better treatment guidance, and personalized therapy [1], [2], [3], [4]. However, existing continuous blood pressure (BP) monitoring solutions still face challenges in balancing medical accuracy and user convenience, particularly in mobile settings. Wearable sensors [5], [6] enable continuous BP monitoring, but prolonged skin contact may lead to issues such as dermatitis [7], [8]. To address this, radar-based approaches [9], [10], [11] have emerged, aiming to eliminate skin contact [12], [13] and estimate BP by analyzing subtle body-surface vibrations caused by blood flow [14], [15].

However, continuous BP estimation remains challenging. Compensatory mechanisms can mask abnormal hemodynamics by making pulse wave amplitude and shape appear normal, leading to inaccurate BP estimates [16]. Furthermore, propagation interference and limited resolution make radar signals more prone to noise than wearable alternatives [16], [17]. As a result, models often degenerate to the mean SBP and DBP of datasets rather than capture dynamic BP when no additional demographic information or calibration is available. This limitation can be partially mitigated by positioning the radar closer to the body to improve signal quality [14], [18], though at the cost of user convenience. At its core, most radar systems, similar to photoplethysmography (PPG)-based methods, rely on pulse wave analysis [6] from a single peripheral site (e.g., wrist or neck), using features such as reflection time and wave amplitude. This oversimplifies the complex, distributed nature of vascular hemodynamics required for accurate BP estimation.

The pursuit of multi-site peripheral arterial wave analysis [15], [19], [20] has emerged as a promising direction to bridge the gap between non-contact sensing and clinical-grade hemodynamic monitoring. This approach is based on estimating pulse transit time (PTT) [21], which differs from single-wave reflection time by measuring the pulse wave travel time between two distinct sites. As the distance of blood flow increases, the measured transit time becomes less sensitive to noise and enables the estimation of wave velocity [22]. Current implementations can be categorized into single-radar multi-point measurements [23], [24] and dual-radar systems [11], [20], [25], [26].

The key challenges of the single-radar approach include signal separation issues and reduced measurement resolution. While beamforming during transmission [23] can enhance resolution, it introduces additional complexity and may cause loss of synchronization between measurement points.

In contrast, directly sensing multi-site pulse waves with distributed radars can accurately measure PTT [21]. However, current studies remain at a preliminary stage, focusing primarily on demonstrating feasibility by fitting statistical patterns rather than addressing the challenges of generalization and robustness.

To address this challenge, we recognize that existing approaches fail to adequately model nonlinear BP dynamics by overlooking key physiological signals related to BP, such as Inter-Beat Intervals (IBI) and vascular transit times. Moreover, current methods often treat respiration-induced fluctuations as

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noise [19], whereas these mechanisms play a crucial role in BP regulation. As a result, most existing works that focus solely on pulse wave features typically require calibration to enhance performance. Meanwhile, end-to-end prediction models that directly use filtered signals as input are prone to over-fitting, as they struggle to generalize due to the relatively small variation in BP.

In this work, we introduce a dual-radar system combined with comprehensive physiological feature analysis to understand better the relationship between BP and various physiological signals extracted from radar measurements. To achieve calibration-free continuous BP estimation, we develop a physiologically-guided neural network that leverages these signals effectively. A key challenge in implementing this approach is identifying optimal measurement positions to ensure consistent vital signal acquisition across dual radars. Conventional signal selection strategies are suboptimal in a dual-radar system, as they fail to account for the coherence between chest and neck radars. Additionally, the inherently lower signal-to-noise ratio (SNR) of the neck radar further increases the challenge of robust signal selection. Therefore, we propose a joint signal selection method to extract robust physiological signals, ensuring that BP-related features are captured reliably with minimal interference.

Our key contributions are three-fold:

1) We present the first contactless system capable of continuous systolic and diastolic BP estimation without requiring per-subject calibration. By integrating spatially distributed vascular dynamics through synchronized dual-radar sensing specifically targeting cardiac mechanical signals from the chest and pulse wave dynamics from the carotid artery, our approach overcomes the resolution and information limitations inherent in single-waveform analysis that plague existing radar-based methods, thereby addressing a critical gap in non-contact hemodynamic monitoring.
2) Our custom-designed dual-radar system with hardware-level synchronization effectively addresses the signal separation and temporal alignment challenges inherent in single-radar multi-point implementations, while incurring minimal additional complexity. To further enhance physiological signal extraction, we propose a joint signal selection algorithm for dual-radar sensing, which improves the robustness and reliability of BP-related feature acquisition. We validate the effectiveness of our system through comprehensive experiments, demonstrating a 7.4%–12% reduction in error compared to single-radar approaches and significantly outperforming baseline methods in continuous BP estimation.
3) We collected over 50 hours of data, spanning multiple seasons, different times of the day, various body postures, and multiple radar devices, ensuring comprehensive coverage. Our physiologically-guided deep learning model demonstrated robust DBP and SBP tracking across varied time periods, radar devices, body postures, and environmental conditions, validating its generalizability and effectiveness in real-world settings.

TABLE I
COMPARISON OF ARTERIAL DIAMETER AND PULSATION AMPLITUDE (MM) [27], [28], [29], [30], [31], [32].

Artery	Diameter	Tissue Depth	Amplitude
Common carotid	6.1-8.1	10.0-42.0	0.40-0.84
Femoral	7.1-10.2	17.0-20.9	0.38-0.46
Brachial	3.1-4.9	6.0-13.4	0.16-0.24
Radial	2.8-3.5	∼2.5	0.02-0.05

II. PRINCIPLES OF BP ESTIMATION

Current cuff-less approaches [16] primarily rely on blood flow dynamics and arterial wall dynamics theories to estimate BP. The first approach analyzes pulse wave characteristics induced by blood flow changes, such as amplitude, time difference, and velocity. This is commonly seen in PPG, one of the most widely used wearable devices, which estimates parameters like the reflective wave transit time of a single pulse wave. The second approach directly estimates blood vessel diameter, as demonstrated in ultrasonic devices [5]. For radar devices, accurately estimating the precise pulse wave key time points or the vessel diameter through penetration of the blood vessel is challenging. Instead, especially at a certain distance, we primarily observe the motion of the skin surface, which reflects the combined effects of blood flow and vascular contraction.

Table I shows the diameter and pulsation amplitude of common arteries. As can be seen, the motion range of the arteries is small, which presents a challenge for radar. Additionally, existing methods for collecting data from the radial artery at the wrist are more susceptible to physiological noise, such as motion artifacts and muscle contractions. Nonetheless, measuring pulse wave velocity (PWV) using multiple arterial measurement sites [33] can improve estimation accuracy. This parameter is widely utilized in clinical practice due to its value in the early diagnosis and management of cardiovascular diseases [34].

A. Measurement Site Selection

Current PWV measurements typically follow three configurations [34]: carotid-femoral PWV (cfPWV), brachial-ankle PWV (baPWV), and brachial-femoral PWV (bfPWV). Among these, cfPWV is regarded as the gold standard for assessing arterial stiffness [35], though it requires highly experienced and trained operators. In contrast, baPWV is a more patient-friendly alternative that is easier for clinicians to implement, while bfPWV remains less extensively studied. Furthermore, these measurements are typically conducted with the subject in a supine position, and achieving high accuracy often depends on the use of ECG [36], [37].

While traditional PWV methods like cfPWV utilize longer arterial paths for optimal clinical assessment of arterial stiffness, Physio-BP's choice of the heart-carotid path is tailored for non-contact, continuous radar-based monitoring. The carotid artery offers a strong, superficial pulsation (as indicated in Table I) readily detectable by radar, and its proximity to the heart simplifies the dual-radar setup and synchronization for PAT estimation. Furthermore, this configuration is amenable

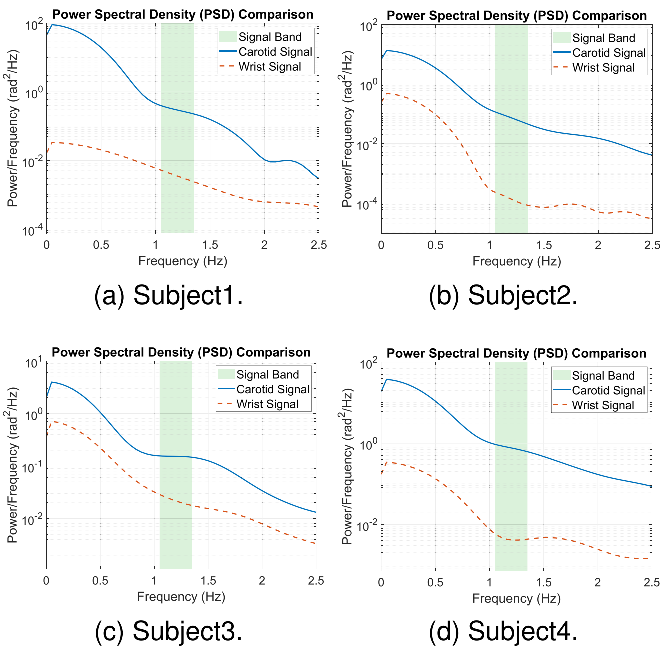


Fig. 1. Comparative power spectral density (PSD) analysis of carotid and wrist radar signals.

to monitoring in common scenarios like sitting or sleeping, whereas thigh measurements, while feasible for some setups, might pose greater challenges for unobtrusive, continuous non-contact sensing with mmWave radar in varied postures.

Building on this foundation, we first select the chest as one of the positions for the dual-radar system. This choice is driven by extensive research on radar-based cardiac monitoring [38], as well as the relatively large amplitude of cardiac motion [39]. Additionally, the chest provides access to the essential ECG modality required for PWV measurement. For arterial measurements, we select the carotid artery, as it is one of the key sites for cfPWV assessment and exhibits the largest vibration amplitude, as shown in Table I. The carotid artery is more accessible for radar-based measurements due to its pronounced surface pulsation and reduced susceptibility to motion artifacts compared to deeper or more mobile arteries such as the femoral or radial arteries.

To address the critical concern regarding potential signal attenuation due to the carotid artery's greater tissue depth, which could challenge our site selection, we performed a comparative spectral analysis of radar signals from the carotid artery (neck) and the radial artery (wrist), a shallower but smaller vessel. As illustrated in Fig. 1 across four representative subjects, the Power Spectral Density (PSD) comparison consistently validates the superiority of the heart-carotid path. In all cases, the Carotid Signal (solid blue line) exhibits power levels that are one to two orders of magnitude higher than the Wrist Signal (dashed red line) within the physiological cardiac band (approximately 0.6 Hz to 2.5 Hz, indicated by the shaded green area). This robust dominance of the carotid signal confirms that its prominent arterial pulsation is sufficient to overcome the attenuation effect of soft tissue, thereby yielding a stronger, more reliable surface vibration required for non-contact radar measurement. This experimental evidence strongly supports our selection of the heart-carotid configuration for optimal continuous hemodynamic monitoring.

Moreover, the chosen heart-carotid artery configuration simplifies the overall radar setup. Crucially, precise hardware-level synchronization between the two radar units is readily achieved through a simple trigger signal connection, obviating the need for complex external synchronization mechanisms or post-hoc data alignment. This inherent simplicity of setup and operation means the subject only needs to sit calmly facing the sensors, without requiring intricate physical alignment procedures. While this heart-carotid configuration proves effective and user-friendly, future work will explore alternative configurations.

B. Heart-Carotid Circulatory

Fig. 2(a) illustrates the hemodynamic mechanism within the heart-carotid artery system. During ventricular systole, the left ventricle contracts and ejects blood into the aorta, creating a pressure wave that distends the arterial walls. The peak SBP correlates with both the stroke volume (SV) and the rate of ventricular ejection [40], [41]. Greater SV and faster ejection velocity produce higher arterial wall tension, thereby elevating SBP. This distension propagates through the elastic arteries as a pulse wave. This non-contact detection of the time delay (Pulse Arrival Time, PAT) between the cardiac ejection (chest) and the carotid pulse (neck) is the core sensing principle of our system, as illustrated in Fig. 2(b).

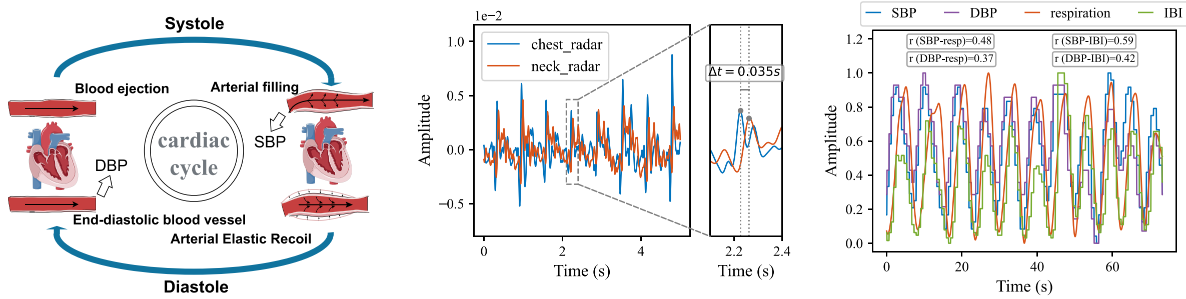
At the end of systole, the closure of the aortic valve marks the transition to diastole. During this relaxation phase, blood continues to flow peripherally due to 1) elastic recoil of the arterial walls and 2) sustained peripheral vascular resistance (PVR). DBP represents the lowest arterial pressure just before the next cardiac cycle, primarily determined by PVR and arterial compliance [40]. Higher vascular resistance impedes diastolic runoff, maintaining elevated DBP.

Based on these physiological underpinnings, we hypothesize that integrating chest radar, which captures signals closer to cardiac ejection, will primarily enhance SBP estimation accuracy, while neck radar, sensing carotid artery dynamics reflective of peripheral resistance and arterial compliance, will predominantly improve DBP estimation.

C. Cardiorespiratory Coupling

Previous studies on discrete BP measurement [42] often exclude respiratory signals, focusing primarily on cardiac-related information. For continuous BP monitoring, previous works [24] have adopted breath-holding protocols to eliminate respiratory interference. However, current medical evidence suggests that spontaneous breathing intrinsically modulates BP through physiological mechanisms [43], [44]. During respiration, thoracic cavity expansion and contraction directly influence cardiac output and venous return, thereby regulating both systolic and diastolic pressure, creating physiologically significant BP oscillations that contain valuable diagnostic information.

In our experimental data, we observed distinct physiological coupling relationships among heartbeat intervals, BP variations, and respiratory patterns, as illustrated in Fig. 2(c). The figure



(a) Blood Pressure Mechanism. (b) Pulse Time Detection. (c) Relationship of BP, Respiration, and IBI.

Fig. 2. Physiological principles of non-contact BP monitoring. (a) Illustrates the cardiac cycle mechanism; (b) shows the non-contact pulse time detection; (c) details the relationships between IBI, respiration, and BP.

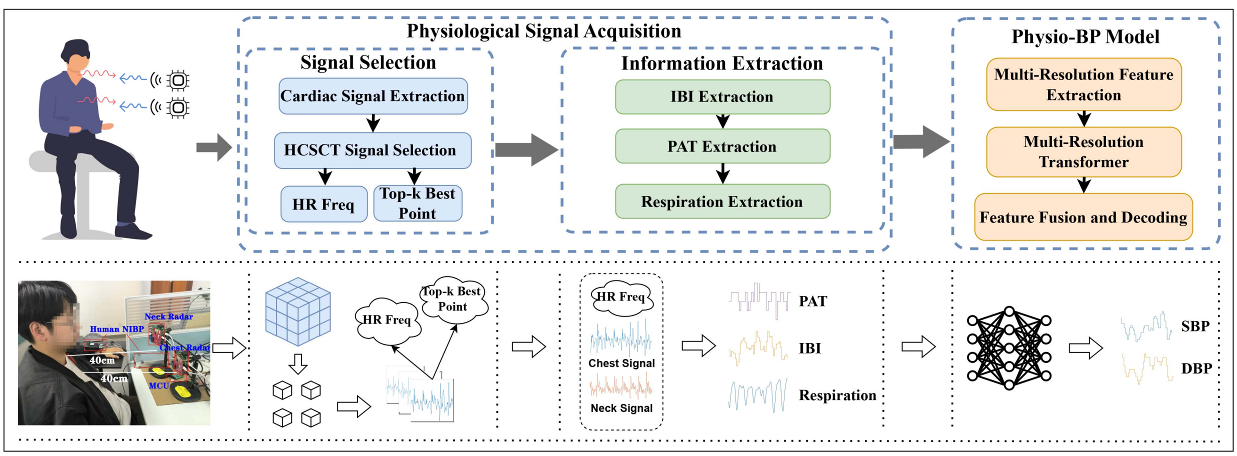


Fig. 3. System design of Physio-BP. The system consists of two main components: (1) Physiological signal acquisition, which extracts physiological information from radar signals, and (2) The physiologically-guided network, which integrates physiological priors into a deep learning framework to enhance blood pressure estimation performance. The bottom section illustrates the real-world setup.

quantifies these couplings, showing clear positive correlations between the respiratory signal and both SBP (r = 0.48) and DBP (r = 0.37). A similar, strong relationship is observed between IBI and both SBP (r = 0.59) and DBP (r = 0.42). This phenomenon, known as respiratory sinus arrhythmia, is a typical example of how respiration influences heart rate. Considering this intrinsic connection between respiration and BP, we believe that incorporating respiratory information can compensate for some inaccuracies in heart rate estimation and provide a macro-trend for BP changes.

III. System Design

Fig. 3 illustrates the overall design of our system. In this section, we first introduce the fundamentals of radar signal processing. We then describe the extraction of physiological information from the dual-radar system. Finally, we present our physiologically-guided network, which incorporates the extracted physiological priors into the deep learning framework to enhance blood pressure estimation.

A. Radar Signal

Current radar-based approaches [10], [18], [23] typically extract pulse wave-related signals by analyzing blood flow dynamics driven by cardiac activity, leveraging features such as PTT for BP estimation [21], [22]. To localize and detect desired physiological signals [45], we employ a Frequency-Modulated Continuous-Wave (FMCW) radar in combination with a 2D antenna array to isolate reflections from each 3D location. With spatial beamforming, we obtain a set of phase signals $\{\phi_c(p,t), p \in \Omega\}$ for the chest radar and $\{\phi_n(p,t), p \in \Omega\}$ for the neck radar.

The phase of the received signal at each location p captures the micro-motion $r_m(t)$ at a distance d over time as follows [46]:

$$\phi_m(t) = 4\pi \frac{r_m(t) + d}{\lambda}, m \in \{c, n\}, \tag{1}$$

where λ is the wavelength of the radar signal, and m represents the different radar types. For the chest radar, $r_c(t)$ mainly includes the displacement of the chest surface caused by breathing and heartbeat. In contrast, for the neck radar, $r_n(t)$ primarily captures the micro-vibrations of the blood vessels resulting from hemodynamic changes.

B. Physiological Signal Acquisition

This module comprises cardiac signal selection and physiological information extraction.

Given the set of phase signals, our goal is first to select the most optimal cardiac signal $V_m(t)$. Based on this selected cardiac signal, we then extract key physiological information: $\mathrm{IBI}_m(t)$, the Inter-Beat Interval extracted from $V_m(t)$; $\mathrm{PAT}(t)$, the pulse arrival time between the neck and heart; and $R_c(t)$, the respiratory signal derived from the chest radar.

However, accurate Inter-Beat Interval (IBI) extraction remains a critical challenge in non-contact hemodynamic monitoring due to the compounded noise introduced by neck arterial dynamics. To address this, we propose the Harmonic-Conscious Synchrosqueezed Cardiac Tracker (HCSCT), which incorporates two key principles: (1) signal enhancement using synchrosqueezed CWT to concentrate cardiac energy, and (2) harmonic validation to ensure IBI consistency across spatial points.

1) Cardiac Signal Selection: Cardiac signals primarily consist of pulse waves from arterial pressure changes with each heartbeat, reflecting vascular expansion and contraction. To extract cardiac vibrations, prior studies apply second-order differentiation to the phase signal $\phi_m(t)$ to suppress respiratory interference, isolating high-frequency cardiac components. We follow this approach and employ a noise-robust second-order differentiation filter [47] for enhanced signal extraction:

$$V_m(t) = \frac{1}{4^{N-1} \cdot h^2}(s * \phi_m)(t),\ m \in \{c, n\}, \tag{2}$$

where h represents the time step, s is a symmetrical filter with length $L = 2N + 1$, satisfying $s[-k] = s[k]$ and the condition $s[N] = 1$. The filter coefficients $s[k]$ follow the recurrence relation:

$$s[k] = \frac{2(N-2)}{N-k} \cdot s[k+1] - \frac{(N+k+2)}{N-k} \cdot s[k+2],\ k \geq 0. \tag{3}$$

In our implementation, we use $L = 43$.

Heart rate detection: For each radar, a simple approach is to select the spatial location with the highest energy as the cardiac-related signal. However, this method is highly susceptible to noise, particularly in neck radar signals. To enhance signal quality and provide a reliable prior for subsequent IBI and PAT estimation, we propose a location selection algorithm. We first identify a small region around the maximum energy position in the chest radar. The continuous wavelet transform (CWT) is applied to each cardiac signal $V_c(t)$ in chest radar as follows:

$$W_c(a, b) = \int_{-\infty}^{\infty} V_c(t) \frac{1}{\sqrt{a}} \psi^* \left(\frac{t-b}{a} \right) dt \tag{4}$$

where $W_c(a, b)$ is the coefficient at scale a and time b, and $\psi(t)$ is the mother wavelet. Then, the Synchrosqueezing Transform (SST) is employed to sharpen the time-frequency resolution of radar signals by reassigning wavelet coefficients to their instantaneous frequencies to concentrate cardiac-related energy and suppress noise interference as follows:

$$S_c(\omega, b) = \sum_a W(a, b) \cdot \delta\left(\omega - \omega_c(a, b)\right), \tag{5}$$

where $\omega_c(a, b)$ is the instantaneous frequency derived from the phase of the wavelet transform.

To obtain the energy spectrum for a given frequency ω, we integrate the energy over time from the time-frequency representation $S_c(\omega, b)$:

$$E_c(\omega) = \sum_b |S_c(\omega, b)|^2, \tag{6}$$

which is then smoothed with the Savitzky-Golay filter. Since the fundamental frequency of the heart rate may not always have the highest energy, we select the two highest energy frequencies, ω_1 and ω_2 , within the range of 0.6Hz to 2.5Hz. If ω_2 is an integer multiple of ω_1, i.e., the harmonic of ω_1, we choose ω_1 as the heart rate frequency. Otherwise, we select the frequency with the higher energy between ω_1 and ω_2.

For all points, we similarly extract all potential heart rate frequencies and then statistically determine the two most frequent ones, ω_a and ω_b. We then check if these two frequencies satisfy the harmonic relationship and neglect the harmonic if it exists. Otherwise, we select the most frequent frequency as the final frequency, denoted as ω_0.

Signal selection: Given ω_0, we search for peak frequency energy within $\omega_0 \pm 0.4$ Hz for each point using the original energy spectrum. We identify each point's peak frequency and energy $(\omega_{\text{peak},i}, E_{\text{peak},i})$ and locate the nearest local minimum $E_{\text{min},i}$ to its right. The ratio $R_i = E_{\text{peak},i}/E_{\text{min},i}$ represents the fundamental frequency clarity.

After sorting these ratios, we select the top k points with highest clarity as our candidate set Ω_{selected}. Fig. 4 illustrates the complete signal processing pipeline from radar data to frequency analysis for heartbeat extraction and point select.

To quantitatively validate the HCSCT algorithm's critical contribution to noise mitigation, we directly compared its Inter-Beat Interval (IBI) estimation accuracy against a conventional peak detection baseline using the same chest radar data. The HCSCT-guided estimation achieved high fidelity, yielding a low mean error (Bias) of -5.04 ms and a high precision (Standard Deviation) of 43.40 ms. In sharp contrast, the baseline method produced significantly poorer results, exhibiting a higher mean error of +21.91 ms and a much larger standard deviation of 112.50 ms. This analysis confirms that HCSCT dramatically enhances IBI precision by over 61%, reducing the standard deviation from 112.50 ms to 43.40 ms compared to the traditional approach.

2) Information Extraction: Given the cardiac signals, we extract complementary information as follows:

IBI: We use the heart rate ω_0 to constrain the cardiac signal for the peak detection and the extraction of the IBI signal. Given the cardiac signal $v_m(t)$, during peak detection, we calculate the time difference between consecutive peaks and select the one with the time difference closest to the expected heart rate frequency as follows:

$$i = \arg\min_i \left| \Delta t_i - \frac{2\pi}{\omega_0} \right|, \tag{7}$$

where Δt_i is the time difference between the potential peak i and the current peak.

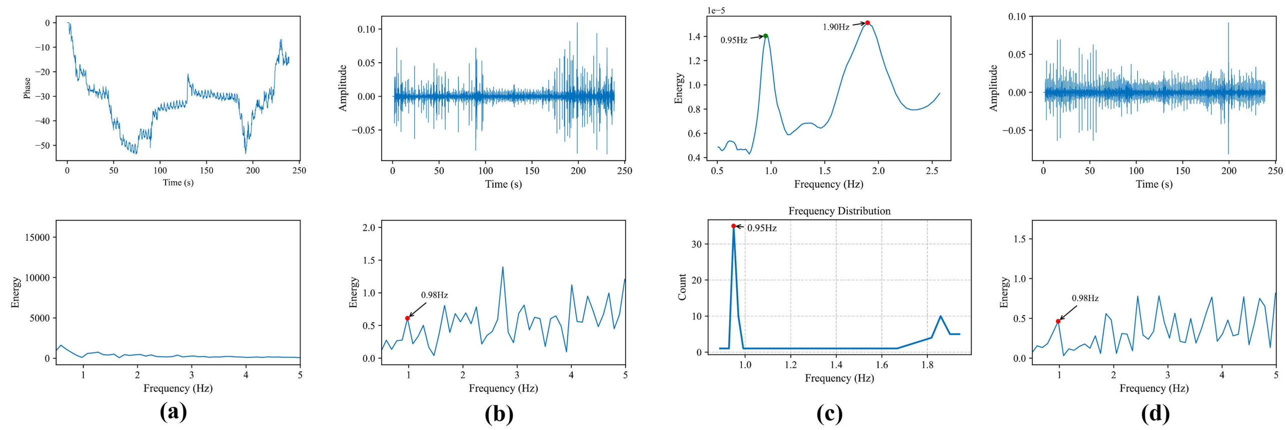


Fig. 4. Pipeline of the harmonic-conscious synchrosqueezed cardiac tracker (HCSCT) using neck radar signals. (a) Phase signal, (b) Cardiac signals, (c) Base frequency detection, and (d) Selected signal.

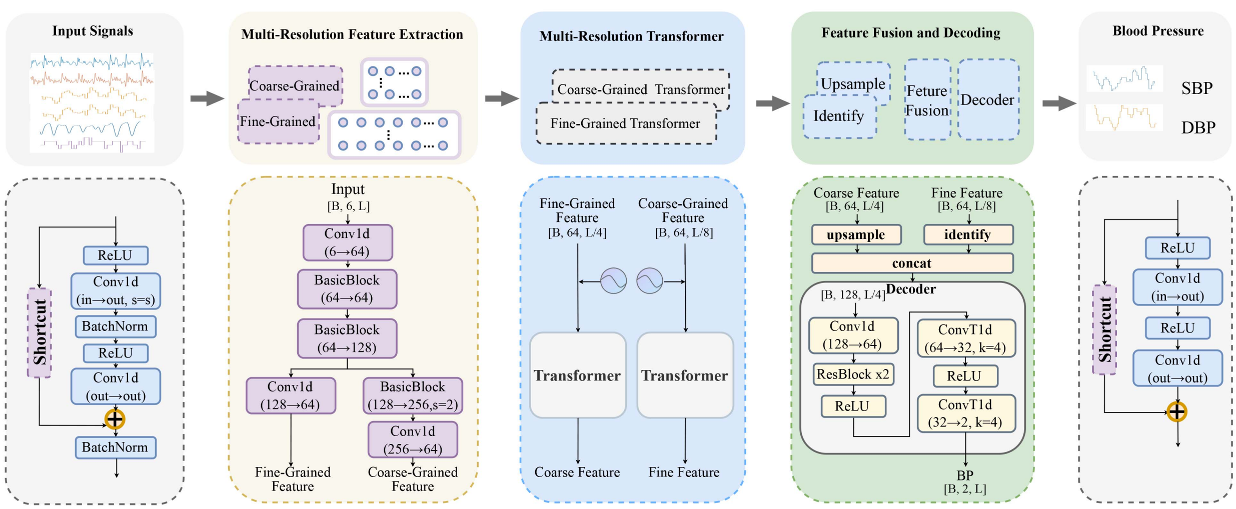


Fig. 5. Model framework. Overview of the Physio-BP model for continuous BP estimation. The framework includes: (1) Multi-Resolution Feature Extractor, which processes input signals into fine- and coarse-grained features; (2) Multi-Resolution Transformer, which models temporal dependencies separately for each scale; (3) Feature Fusion Layer, which aligns and integrates features; and (4) Decoder, which reconstructs continuous SBP and DBP predictions.

PAT: The PAT is calculated as the difference between the IBI signals from the chest and neck radars:

$$\mathrm{PAT}(t) = \mathrm{IBI}_c(t) - \mathrm{IBI}_n(t), \tag{8}$$

where $\mathrm{IBI}_c(t)$ and $\mathrm{IBI}_n(t)$ represent the IBI signals from the chest and neck radars, respectively.

Respiration: We apply filtering to $\phi_m(t)$ to obtain the respiratory signal:

$$R_m(t) = \mathscr{F}(\phi_m(t)), \tag{9}$$

where $\mathscr{F}(\cdot)$ represents the low-pass filter.

C. Model Architecture

In this section, we introduce Physio-BP, a physiologically-guided network for continuous BP estimation. As illustrated in Fig. 5, Physio-BP integrates multi-resolution feature extraction, transformer-based temporal modeling, and feature fusion to improve prediction accuracy.

Our model utilizes a dual-radar system (neck and chest) to capture cardiovascular dynamics. The input is a multivariate time series:

$$x(t) = [V_c, V_n, IBI_c, IBI_n, PAT, R_c](t) \in \mathbb{R}^{6\times T}, \tag{10}$$

where T is the time series length. This is a sequence-to-sequence prediction problem, where the model estimates BP at each time step:

$$y(t) = [\mathrm{SBP}(t), \mathrm{DBP}(t)] \in \mathbb{R}^{2\times T}. \tag{11}$$

The model learns temporal dependencies to predict SBP and DBP over time.

Our model's multi-resolution architecture is explicitly guided by the physiological principles established in Section II. Continuous BP is a composite signal, simultaneously reflecting distinct temporal scales of hemodynamic activity. These comprise fine-grained dynamics, such as rapid, beat-to-beat cardiac events (V_c, V_n, PAT, IBI), and coarse-grained dynamics, such as the slower modulatory macro-trends driven by cardiorespiratory coupling (R_c). A conventional single-scale architecture would be ill-equipped to concurrently optimize for these fast and slow processes. Physio-BP's multi-resolution framework is thus designed to explicitly disentangle and model these distinct physiological drivers in parallel, thereby enhancing the estimation accuracy for both SBP and DBP.

1) Multi-Resolution Feature Extraction: We employ a multi-resolution approach to capture physiological features across different temporal scales. The low-level analysis identifies subtle temporal relationships between neck and chest signals associated with systolic pressure, while the high-level analysis models respiratory sinus arrhythmia and thoracic pressure variations that influence diastolic pressure:

$$\begin{aligned} v_r &= \mathscr{F}_r(x) \in \mathbb{R}^{D\times T}, \\ v_l &= \mathscr{F}_l(v_r) \in \mathbb{R}^{D\times \frac{T}{4}}, \\ v_h &= \mathscr{F}_h(v_r) \in \mathbb{R}^{D\times \frac{T}{8}}, \end{aligned} \tag{12}$$

where $\mathscr{F}_r$ extracts initial features from x, $\mathscr{F}_l$ captures lowlevel temporal dynamics relevant to systolic pressure estimation, and $\mathscr{F}_h$ extracts high-level features that model respiratory and thoracic influences on diastolic pressure.

2) Multi-Resolution Transformer: Since the self-attention mechanism in Transformers is permutation-invariant, it does not inherently capture the temporal order of the input sequence. To address this, we apply positional encodings (PE) to the fine-grained (v_l) and coarse-grained (v_h) feature sequences before they are fed into the Transformer encoders.

First, the positional encodings ($PE_l \in \mathbb{R}^{D\times \frac{T}{4}}$ and $PE_h \in \mathbb{R}^{D\times \frac{T}{8}}$) are added to the feature sequences:

$$\begin{aligned} v'_l &= v_l + PE_l \\ v'_h &= v_h + PE_h \end{aligned} \tag{13}$$

To preserve distinct temporal characteristics across different resolutions, we employ separate Transformer encoders for each scale:

$$z_l = \mathscr{G}_l({v'}_l) \in \mathbb{R}^{D\times \frac{T}{4}}, z_h = \mathscr{G}_h({v'}_h) \in \mathbb{R}^{D\times \frac{T}{8}}. \tag{14}$$

Each encoder leverages multi-head self-attention to establish global consistency across time while mitigating potential information loss due to missing frames in radar signals.

3) Feature Fusion and Decoding: The feature fusion layer aligns and integrates multi-resolution features while preserving their complementary information:

$$z_o = \text{Concat}(\text{Upsample}(z_h), z_l) \in \mathbb{R}^{2D\times \frac{T}{4}}. \tag{15}$$

This progressive upsampling approach retains physiologically relevant details from both radar sources, enabling continuous predictions of SBP and DBP.

The decoder reconstructs continuous BP tracing:

$$\hat{y} = \mathscr{D}(z_o) \in \mathbb{R}^{2\times T} \tag{16}$$

where $\mathscr{D}(\cdot)$ is the classifier.

TABLE II
RADAR CONFIGURATION

Parameter	Value
Start Frequency	77 GHz
Chirp Slope	99.987 MHz/μs
Chirp Duration	40 μs
Chirp Repetition Rate	200 Hz
Idle Time	80 μs
ADC Samples	256
ADC Sampling Rate	8000 kbps
RX Gain	48 dB

4) Objective Function: We employ the Mean Squared Error (MSE) loss as the objective function to optimize BP prediction:

$$\ell = \frac{1}{2T}\sum_{t=1}^{T} |\hat{y}_t - y_t|^2 . \tag{17}$$

By combining neck and chest measurements, Physio-BP reduces sensitivity to anatomical variations. The integration of IBI and PAT provides complementary timing references, improving estimation accuracy. Additionally, modeling respiratory patterns helps account for respiratory-driven BP variations. Unlike traditional spot-check methods, Physio-BP enables continuous monitoring, offering a more comprehensive assessment across different physiological states and conditions.

IV. EXPERIMENT SETUP

A. Data Collection

1) Dual-Radar Configuration: We position two mmWave radars (TI AWR1843-BOOST) closely together to form a dual-radar system for data acquisition. Placed 40 cm in front of the subject, the radars are directed at the chest and neck regions, respectively. Both radars operate with identical configurations in Table II and are synchronized by a microcontroller with a 50 μs timing offset between their chirp trigger times to prevent mutual interference. Synchronization is achieved by connecting the timing signals to each radar's `SYNC_IN` port. This dual-radar configuration allows us to acquire complementary cardiovascular dynamics from different anatomical locations, ensuring precise temporal alignment for accurate blood pressure estimation.

This co-located dual-radar setup ensures consistent data collection while capturing complementary cardiovascular dynamics from two physiologically distinct regions. By maintaining precise temporal alignment, this configuration enhances the accuracy of BP estimation.

2) Participant and Data Acquisition Protocol: We recruited 56 participants (44 males, 12 females) for this study to collect data under diverse environmental conditions. The final dataset comprises approximately 50 hours of synchronized radar (monitoring chest and neck regions) and reference BP recordings.

The reference device used in this study is based on the Finapres (volume-clamp) method, providing continuous, beat-to-beat ground-truth blood pressure measurements. The detailed specifications are listed in Table III. During the acquisition of these reference BP values, participants adhered to standard

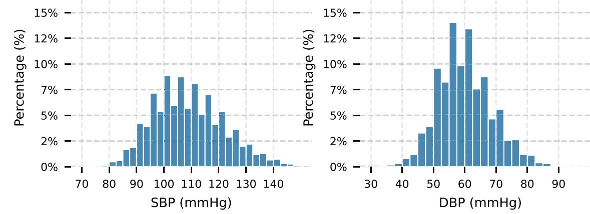
TABLE III
HUMAN NIBP NANO (ADINSTRUMENTS, INL382) SPECIFICATIONS

Parameter	Specification
Device Name	Human NIBP Nano
Model	ADInstruments, INL382
Technology	Finapres (volume-clamp)
Measurement	Continuous, beat-to-beat
Interface	Dual-finger cuff
BP Accuracy	1% of full scale (max. 3 mmHg)
Temporal Resolution	10 ms (max, non-accumulating)

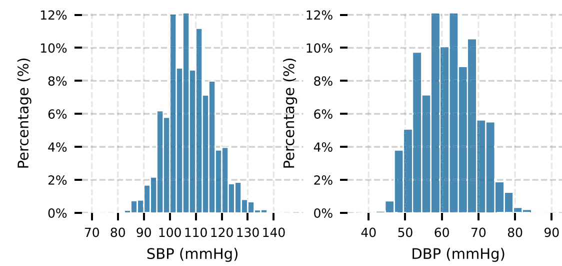
clinical guidelines, which included positioning their arm at heart level and pre-warming the BP device. Throughout both reference BP measurements and subsequent radar data collection for the Physio-BP system, participants were asked to remain relatively still in a comfortable seated or supine position, minimizing gross body movements. Natural respiratory movements and minor postural adjustments were permitted. Each data collection session lasted 3 to 5 minutes, with multiple sessions per participant contributing to an average of approximately 0.9 hours of data per person.

It is important to emphasize that the arm-at-heart-level positioning was a procedural requirement solely for the reference BP monitor. The Physio-BP system, operating via radar sensing of the chest and neck regions, does not depend on this specific arm posture. However, maintaining consistent subject positioning relative to the Physio-BP radars (i.e., stable orientation of the chest and neck towards the sensors) is generally beneficial for optimal signal quality from our system.

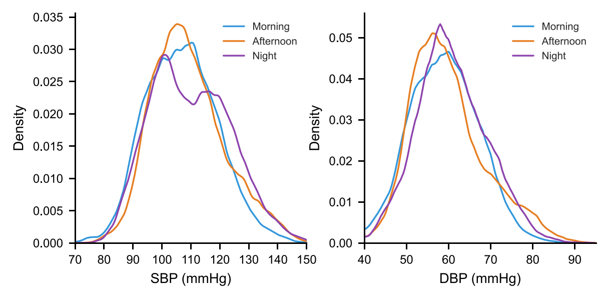
The overall distribution of SBP and DBP measurements across train and validation participants is shown in Fig. 6(a), and the distribution for test participants is shown in Fig. 6(b). Additionally, Fig. 6(c) illustrates the normalized blood pressure distributions across morning, afternoon, and night, highlighting temporal variations in the participants' blood pressure profiles.



(a) Distribution of systolic and diastolic blood pressure values across train and valid participants.



(b) Distribution of systolic and diastolic blood pressure values across test participants.



(c) Blood pressure distributions across different time periods.

Fig. 6. Statistical analysis of blood pressure measurements, showing overall distribution and temporal variations.

B. Implementation Details

Our Physio-BP model, a Transformer-based architecture detailed in Section III-C, was trained for 200 epochs using the MSE loss function. We employed the AdamW optimizer with a learning rate of 5×10^{-4}. The Transformer architecture utilized 4 attention heads, a hidden dimension size of 64, and a dropout rate of 0.1 to mitigate overfitting. Training was conducted with a batch size of 256. Throughout the training process, model performance was monitored on a dedicated validation set, and the model achieving the best validation performance was selected for final evaluation on the test set.

1) Dataset and Participant Protocol: The study protocol was approved by our institute's ethics committee, and the final dataset comprised data collected from 56 participants who provided informed consent. During data collection for ground truth labeling with a reference BP monitor, participants were instructed to position their arm at the same level as their heart, adhering to standard guidelines for clinical BP measurement to ensure the accuracy of the reference values. Participants were asked to remain relatively still in a comfortable seated or supine position, minimizing gross body movements, although natural respiratory movements and minor postural adjustments were permitted. It is important to note that while this arm-heart level positioning was crucial for obtaining accurate reference BP data using the Non-Invasive Blood Pressure (NIBP) monitor, the Physio-BP system's radar-based sensing of the chest and neck does not inherently require this specific arm posture. However, consistent subject positioning relative to the radars is generally beneficial for optimal signal quality.

2) Data Splitting and Evaluation Strategy: For robust model training and evaluation, the 56-participant dataset was partitioned to ensure strict subject-wise separation: data from 40 participants were allocated for training, 8 participants for validation, and the remaining 8 unseen participants formed the independent test set. All primary end-to-end performance results, including the baseline comparisons presented in Table IV, are reported on this independent test set unless explicitly stated otherwise.

For the robustness analyses detailed in Table VI (Robust Results across Different Scenarios):

TABLE IV
COMPARISON WITH BASELINE METHODS FOR BLOOD PRESSURE ESTIMATION. C AND N DENOTE RADAR PLACEMENT AT THE CHEST AND NECK, RESPECTIVELY. (P) AND (I) INDICATE PHASE AND I-CHANNEL SIGNAL EXTRACTION METHODS.

Method	Radar	SBP						DBP					
		ME	MAE ↓	STD ↓	MAD ↓	MAPD ↓	RMSE ↓	ME	MAE ↓	STD ↓	MAD ↓	MAPD ↓	RMSE ↓
Ours	N+C	-0.33	**7.32**	**5.77**	**6.43**	**5.99%**	**9.32**	-0.17	**5.69**	**4.19**	**5.32**	**8.79%**	**7.06**
WaveBP	C	-7.33	11.21	12.10	9.60	8.84%	13.61	4.66	8.38	8.94	7.77	13.83%	9.95
	N	-3.05	13.75	16.35	11.51	10.64%	15.60	3.63	8.15	9.43	7.72	13.94%	10.11
mmBP	C	0.03	7.77	9.68	7.27	6.84%	9.68	1.74	7.63	9.25	7.61	12.55%	9.25
	N	-0.15	8.61	10.40	7.52	7.09%	10.40	1.44	7.50	9.25	7.35	12.14%	9.25
BP^3	C (P)	-3.80	8.85	7.14	8.00	7.24%	11.37	-4.49	8.81	5.12	8.56	13.86%	10.19
	C (I)	1.06	7.46	5.86	7.41	7.11%	9.49	-2.04	8.63	4.84	8.55	14.45%	9.89
	N (P)	-0.95	8.33	6.81	7.24	6.71%	10.76	-3.13	8.19	5.13	7.95	13.14%	9.67
	N (I)	-0.35	8.30	6.78	7.23	6.74%	10.72	-2.75	8.04	5.16	7.80	12.97%	9.55
caPTT	N+C	3.03	8.10	6.23	7.60	7.07%	10.47	-1.88	6.32	4.24	7.29	12.65%	10.48
RF-BP	C	-6.03	8.19	6.58	7.00	6.33%	9.19	-5.43	8.16	5.42	7.94	12.29%	9.47
	N	5.02	8.39	5.65	8.03	7.22%	10.51	1.36	6.91	4.57	8.11	12.65%	9.80

TABLE V
ABLATION STUDY RESULTS SHOWING THE IMPACT OF DIFFERENT COMPONENTS ON BP ESTIMATION PERFORMANCE

Configuration	SBP						DBP					
	ME	MAE ↓	STD ↓	MAD ↓	MAPD ↓	RMSE ↓	ME	MAE ↓	STD ↓	MAD ↓	MAPD ↓	RMSE ↓
Full data	-0.33	**7.32**	**5.77**	**6.43**	**5.99%**	**9.32**	-0.17	**5.69**	**4.19**	**5.32**	**8.79%**	**7.06**
w/o diff data	2.02	7.71	5.82	6.77	6.43%	9.66	0.31	5.83	4.27	5.48	9.16%	7.23
w/o neck data	1.54	7.91	6.04	7.11	6.71%	9.95	0.81	6.42	4.50	6.11	10.26%	7.84
w/o chest data	2.51	7.91	5.88	6.87	6.54%	9.86	0.07	6.11	4.33	5.77	9.59%	7.49
w/o resp.	2.36	7.81	5.66	6.90	6.54%	9.64	-0.28	5.85	4.19	5.52	9.10%	7.19
Only diff data	4.09	9.00	6.61	7.89	7.54%	11.17	-1.40	6.70	4.95	6.30	10.22%	8.33
w/o neck data & resp.	1.90	7.84	5.68	6.89	6.50%	9.68	0.19	6.01	4.20	5.69	9.42%	7.33
w/o resp. & PT	2.18	8.60	6.60	8.25	7.60%	11.20	0.66	6.96	4.86	5.99	9.94%	7.98
w/o select	2.89	8.14	5.81	7.28	6.91%	10.01	0.04	6.41	4.40	6.13	10.09%	7.78

TABLE VI
ROBUST RESULTS ACROSS DIFFERENT SCENARIOS

Scenario	SBP						DBP					
	ME	MAE ↓	STD ↓	MAD ↓	MAPD ↓	RMSE ↓	ME	MAE ↓	STD ↓	MAD ↓	MAPD ↓	RMSE ↓
Cross-season (Jul→Dec)	-0.16	7.94	6.16	6.91	6.48%	10.05	0.22	7.10	4.61	6.84	11.41%	8.47
Morning	3.27	8.65	6.28	7.94	7.63%	10.69	0.10	6.46	4.16	6.23	10.36%	7.69
Afternoon	2.21	6.92	4.97	5.77	5.41%	8.52	0.17	5.16	4.03	4.67	7.81%	6.55
Night	3.30	8.39	5.96	7.43	7.13%	10.29	0.51	6.77	4.45	6.48	10.82%	8.11
Lying position	-4.47	8.29	6.23	7.91	6.84%	10.37	-1.98	4.07	3.19	3.78	5.94%	5.17
New device (AWR6843AOP)	4.22	8.68	5.84	7.48	7.16%	10.46	2.04	4.76	3.48	4.25	7.25%	5.89

- The 'Cross-season', 'Lying position', and 'New device' evaluations involved training the model on the original 40-participant training set and testing its generalization on distinct, newly collected datasets specific to these conditions. These test datasets featured participants entirely unseen during the training, validation, or main test phases.
- The 'Morning', 'Afternoon', and 'Night' evaluations were performed by segmenting the data from the 8 participants in the main independent test set according to the time of day of data collection.

Crucially, all robustness evaluations were conducted in a cross-subject manner, maintaining complete subject independence between training and testing phases for all reported results.

3) Cross-Device Evaluation Protocol: In the 'New device' robustness test, we compared performance using the original AWR1843BOOST radar and the AWR6843AOP radar. The AWR1843BOOST was configured with a start frequency of 77 GHz, as detailed in Table II. The AWR6843AOP, operating in the 60-64 GHz band, was configured with a start frequency of 60 GHz. Apart from this necessary difference in the start frequency due to the distinct operating bands of the radar platforms, all other FMCW parameters were kept identical for both radar setups. This ensured that the comparison primarily evaluated the adaptability of our signal processing and BP estimation algorithms to signals from different frequency bands and radar front-ends, while maintaining consistency in other waveform characteristics.

C. Evaluation Metrics

The common metrics we used are Mean Error (ME), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). In addition to commonly used evaluation metrics, we incorporate the IEEE standards for wearable, cuffless blood pressure monitoring devices [48] as existing cuff-based metrics can sometimes lead to misleading conclusions [49].

Let p_i be the test device measurement, and r_i be the average of the two reference measurements taken before and after the device measurement, We use the following additional metrics:

Mean Absolute Difference (MAD):

$$\mathrm{MAD} = \frac{1}{n}\sum_{i=1}^{n}|p_i - r_i|. \tag{18}$$

Mean Absolute Percentage Difference (MAPD):

$$\mathrm{MAPD} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{100(p_i - r_i)}{r_i}\right|. \tag{19}$$

V. Result

A. BP Estimation

1) Overall Performance: We first compare our proposed method with several established radar-based blood pressure estimation approaches. The experimental results in Table IV show that our approach achieves significantly improved performance, with SBP/DBP mean errors of $-0.33/-0.17$ mmHg and MAEs of 7.32/5.69 mmHg. These results highlight the effectiveness of our method in enhancing the accuracy of blood pressure estimation.

Beyond conventional metrics, we also assess performance using MAD, a key criterion specified by IEEE Standard 1708-2014 [48] for non-invasive cuffless blood pressure devices. This standard mandates MAD values below 7 mmHg for both SBP and DBP. Our dual-radar approach achieves MAD values of 6.43 mmHg for SBP and 5.32 mmHg for DBP, successfully meeting this critical clinical benchmark.

2) Performance With Different Radar: Given the reliance on single-sensor configurations in existing methods, we evaluate baseline approaches across different measurement positions to assess their sensitivity to positional variations and consistency in effectiveness.

WaveBP exhibits position-dependent performance, with chest measurements (SBP/DBP MAE = 11.21/8.38 mmHg) differing from neck measurements (SBP/DBP MAE = 13.75/8.15 mmHg). In contrast, mmBP demonstrates more consistent performance across positions, yielding chest (SBP/DBP MAE = 7.77/7.63 mmHg) and neck (SBP/DBP MAE = 8.61/7.50 mmHg) measurements. For BP^3, we conduct a detailed evaluation comparing both I-channel and phase information, as its original implementation utilizes a wrist-worn sensor in close proximity, whereas our setup operates at greater distances. The results indicate that I-channel processing at the chest position (SBP/DBP MAE = 7.46/8.63 mmHg) outperforms phase-based approaches for SBP estimation. RF-BP displays position-specific characteristics, with chest measurements (SBP/DBP MAE = 8.19/8.16 mmHg) differing from neck measurements (SBP/DBP MAE = 8.39/6.91 mmHg). These findings highlight the influence of sensor placement on prediction accuracy and emphasize the importance of optimizing measurement positions for improved performance.

This analysis reveals a consistent pattern across most methods: SBP measurements are more accurate at the chest, while DBP measurements perform better at the neck. This trend holds for most baselines. The chest's proximity to the heart enhances SBP estimation by capturing stronger cardiac contraction signals, while the neck, with its prominent carotid artery, better reflects DBP-related arterial compliance and wave reflection. These findings underscore the need for position-specific optimization in radar-based BP monitoring and suggest that hybrid approaches leveraging multiple sites could enhance overall accuracy.

3) BP Tracking: We assess the method's ability to track dynamic blood pressure variations over time. As shown in Fig. 7(a), our approach achieves effective continuous BP tracking, with correlation coefficients of 0.725 for SBP and 0.740 for DBP. The predicted BP waveforms closely align with the ground truth, accurately capturing both rhythmic fluctuations and sudden changes in blood pressure. Error analysis for this sample indicates minimal bias, with mean errors of 2.73 mmHg for SBP and -0.27 mmHg for DBP and standard deviations of 4.16 mmHg and 1.51 mmHg, respectively.

Additionally, we evaluate WaveBP for chest and neck measurements, as shown in Fig. 7(b) and (c), further highlighting the superior performance of our approach.

4) CDF Analysis of Continuous BP Tracking: We evaluated dynamic blood pressure tracking using Correlation Coefficient Cumulative Distribution Function (CDF) with 5-second windows that align with respiratory cycles, balancing temporal resolution and statistical reliability.

Fig. 8(a) demonstrates our method's superior SBP tracking performance with a median correlation of 0.747 versus baseline methods' 0.217. The CDF curves show approximately 60% of our measurements exceed 0.6 correlation, while baseline methods have 35-45% near-zero or negative values, indicating their failure to capture respiratory-related SBP fluctuations.

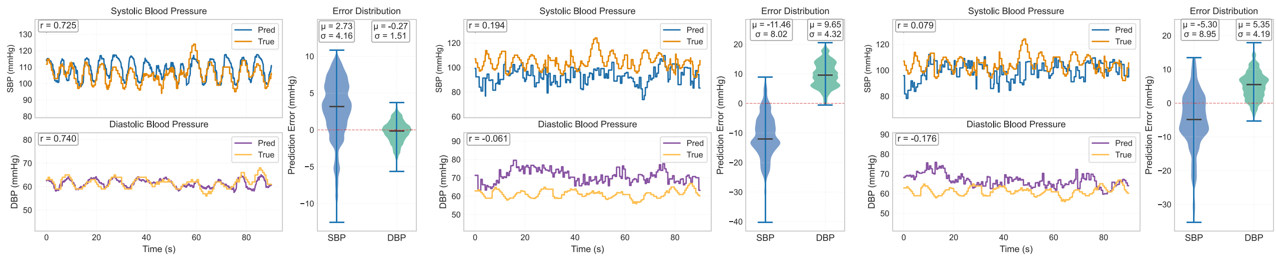
Fig. 8(b) shows that for DBP tracking, despite its greater challenges, our method achieves a median correlation of 0.480 with nearly 50% of measurements exceeding 0.5. Baseline approaches perform substantially worse (median correlation 0.072), failing to capture meaningful diastolic pressure changes.

The distinct separation between the CDF curves highlights our method's superior ability to capture subtle diastolic pressure variations during respiratory cycles.

B. Ablation Study

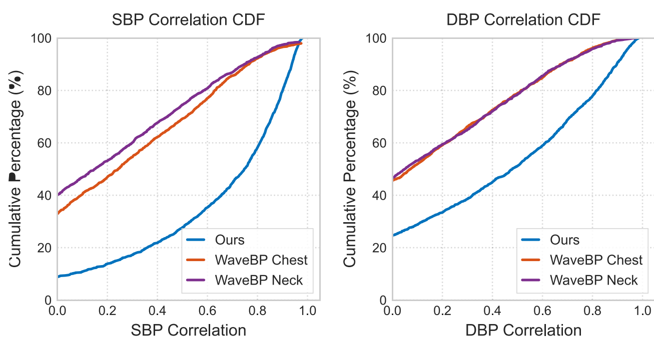
To quantify the contributions of different physiological components and radar data sources, we conducted an ablation study, summarized in Table V and as shown in Fig 9(a) and Fig 9(b).

Our ablation study reveals that removing chest radar data increases SBP MAE by 0.59 mmHg (8.1%), while excluding neck radar data raises DBP MAE by 0.73 mmHg



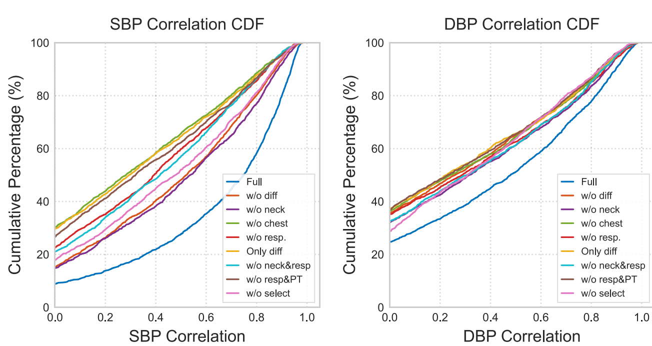
(a) Ours (b) WaveBP (Chest) (c) WaveBP (Neck)

Fig. 7. Comparison of results from our method, WaveBP Chest, and WaveBP Neck.



(a) SBP correlation CDF. (b) DBP correlation CDF.

Fig. 8. SBP and DBP correlation CDF analysis.

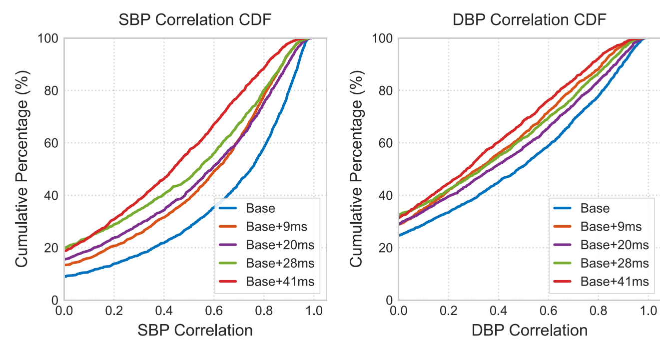


(a) SBP correlation CDF. (b) DBP correlation CDF.

Fig. 9. SBP and DBP ablation study correlation CDF analysis.

(12.8%). These findings corroborate our hypothesis and the position-specific observations from comparative experiments, reinforcing established cardiovascular physiology [40]. Specifically, measurements from the thoracic region, capturing cardiac ejection dynamics, are more influential for SBP estimation. Conversely, carotid artery measurements from the neck region, reflecting arterial compliance and wave reflection characteristics, are more crucial for DBP determination.

The radar second-order differential data enhances tracking accuracy, as its removal raises SBP/DBP MAE by 0.39/0.14 mmHg (5.3%/2.5%). However, relying solely on differential data significantly degrades performance, increasing SBP/DBP MAE by



(a) SBP correlation CDF. (b) DBP correlation CDF.

Fig. 10. SBP and DBP ablation study correlation CDF analysis with different ibi error.

1.68/1.01 mmHg (23.0%/17.8%). This underscores the necessity of integrating differential features with direct physiological measurements to capture complex cardiovascular dynamics effectively.

The adaptive point selection algorithm is crucial, as its removal increases SBP/DBP MAE by 11.2%/12.7%. This underscores its role in dynamically identifying optimal signal locations and addressing individual variability without manual tuning. These results validate our multi-position, multi-signal approach and quantify the benefits of position-specific optimization in radar-based BP monitoring.

We also assessed the system's sensitivity to Inter-Beat Interval (IBI) measurement errors. Fig. 10(a) and (b) show the Cumulative Distribution Functions (CDFs) for SBP and DBP correlation, respectively. The "Base" curve reflects performance using IBI data derived from our proposed method. Subsequent curves ("Base+9 ms" to "Base+41 ms") illustrate the impact of adding incremental MAE of 9 ms to 41 ms to this baseline IBI.

As IBI MAE increases, the CDF curves for both SBP and DBP correlation shift leftwards, indicating a clear degradation in prediction correlation. This demonstrates that higher inaccuracies in our method-derived IBI lead to poorer SBP and DBP prediction performance, likely resulting in increased prediction MAE. These findings highlight the importance of accurate IBI estimation for reliable blood pressure monitoring.

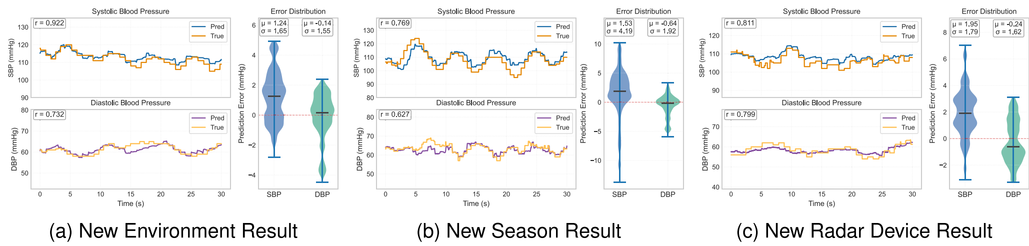


Fig. 11. Comparison across different conditions: (a) New environment, (b) New season, (c) New radar device.

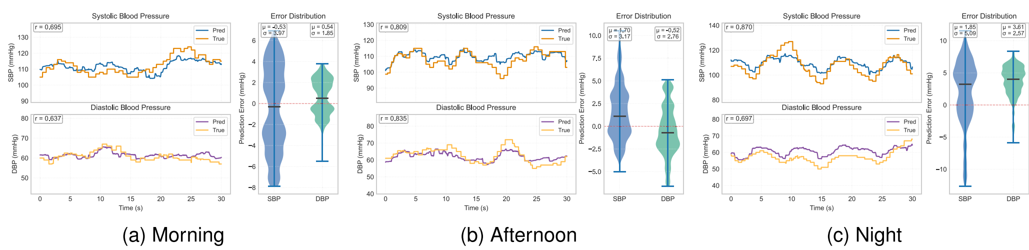


Fig. 12. Comparison of results at different times of the day: (a) Morning, (b) Afternoon, (c) Night.

These findings affirm the effectiveness of our multi-position, multi-signal approach and provide quantitative support for position-specific optimization in radar-based blood pressure monitoring.

C. Robust Analysis

1) New Environment Impact: To evaluate robustness and generalizability, we tested our method in a new environment with 5 previously unseen subjects measured in a lying position. Fig. 11(a) shows our system maintains excellent performance in this challenging scenario. The overall MAE for this new environment was 8.29 mmHg for SBP and 4.07 mmHg for DBP. These preliminary results in a changed environment suggest potential for adaptation to different environmental conditions.

2) New Season Impact: We conducted a challenging cross-seasonal evaluation by training our system on data collected in July and testing on December data. This assessment examines robustness to significant seasonal variations including changes in clothing thickness (winter clothing being substantially thicker), ambient temperature differences, and potential physiological adaptations to different seasons. Fig. 11(b) shows an example of this cross-seasonal performance. The overall MAE across all cross-seasonal tests was 7.94 mmHg for SBP and 7.10 mmHg for DBP. These initial cross-seasonal results provide insights into the system's potential adaptability to different seasonal conditions.

3) New Radar Device Impact: We validated our approach's robustness across different radar platforms by transitioning from the larger AWR1843BOOST to the compact AWR6843AOP. Testing on two previously unseen subjects with the new hardware achieved MAE of 8.68 mmHg (SBP) and 4.76 mmHg (DBP), comparable to our primary system. Fig. 11(c) illustrates this cross-device performance, confirming our method captures fundamental physiological dynamics rather than hardware-specific features, supporting potential miniaturization.

4) Different Periods Impact: We evaluated our system at different times of day (morning, afternoon, and evening) to assess its accuracy during typical blood pressure variations that occur with contemporary lifestyles and irregular schedules. Fig. 12(a), (b), and (c) demonstrate the cross-temporal performance. The system achieved MAEs of 8.65/6.46 mmHg (SBP/DBP) in the morning, 6.92/5.16 mmHg in the afternoon, and 8.39/6.77 mmHg in the evening. These results suggest promising consistency across different times of day, indicating potential for practical daily monitoring applications that accommodate modern variable routines.

5) Individual Diurnal Variation Assessment: We evaluated our system's ability to track intra-individual blood pressure variations by testing on a subject with typical diurnal patterns. This individual exhibited the natural 10 mmHg systolic pressure decline from morning to evening as shown in Fig. 13. The results demonstrate our system's potential to detect physiologically meaningful relative changes within an individual's normal range—a capability particularly valuable for continuous monitoring applications where tracking trends over time can provide important clinical insights beyond single measurements.

6) Different Radar Placements Impact: To address the critical concern of robustness to subject position, we conducted a generalization test. This evaluation posed a dual-domain shift for the model, which was: (1) Trained exclusively on 40cm(0°) data from the training set, and (2) Tested on data from 6 entirely new participants, who were completely unseen by the model. The model was not retrained or fine-tuned in any way. The results of this rigorous cross-subject and cross-position test are presented in Table VII.

TABLE VII
ROBUST RESULTS ACROSS DIFFERENT RADAR PLACEMENTS

Placement	SBP						DBP					
	ME	MAE ↓	STD ↓	MAD ↓	MAPD ↓	RMSE ↓	ME	MAE ↓	STD ↓	MAD ↓	MAPD ↓	RMSE ↓
(a) Distance variation (0° orientation)												
20cm	4.03	8.90	6.14	8.00	8.42%	10.81	6.78	7.47	4.52	7.23	14.23%	8.73
40cm	-0.55	7.03	5.32	6.21	6.02%	8.82	2.95	4.78	4.11	4.38	8.11%	6.31
60cm	0.28	9.83	6.91	8.96	5.41%	8.91	5.67	8.37	5.92	8.07	15.93%	10.25
(b) Angle variation (fixed 40 cm distance)												
+30°	1.92	8.32	6.36	7.43	7.64%	10.47	4.69	6.52	5.01	6.12	11.72%	8.22
-30°	0.35	7.78	5.72	6.58	6.49%	9.65	6.53	8.06	5.98	7.63	14.68%	10.04

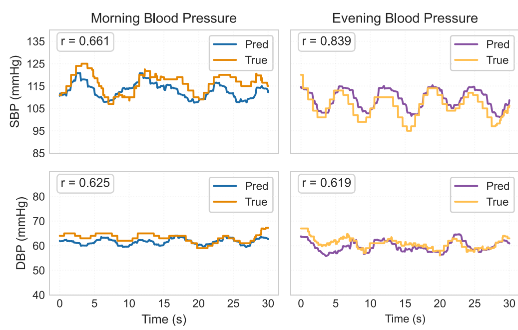


Fig. 13. Tracking of diurnal blood pressure variations in a single individual. The figure compares morning (left) and evening (right) measurements. The system accurately captures the physiological 10 mmHg decline in systolic blood pressure from morning (110-120 mmHg) to evening (100-110 mmHg).

The system achieves its optimal baseline performance on the 40 cm(0°) configuration (SBP/DBP MAE: 7.03/4.78 mmHg). When faced with the dual challenge of unseen subjects and unseen positions, the model exhibits a graceful and moderate degradation in accuracy.

For distance variation, the SBP MAE moderately increased to 8.90 mmHg at 20 cm and 9.83 mmHg at 60 cm. The more significant drop at 60 cm is an expected physical limitation, attributable to signal attenuation and a lower SNR at greater distances. For angular variation, deviations of $\pm 30°$ also resulted in a well-contained performance drop (SBP MAE of 7.78-8.32 mmHg).

Considering that this was a strict, zero-shot evaluation on new subjects and new positions, the ability to maintain MAE values in this range without any retraining strongly confirms the practical robustness and generalization capability of our system.

7) Impact of Body Slight Movements: To evaluate the system's robustness against common motion artifacts, we collected a new dataset from 5 previously unseen participants. During data acquisition, subjects were permitted to perform slight body movements, such as using a smartphone or slight leg shaking, while gross body movements (e.g., large posture shifts or standing up) were excluded.

Under this slight motion interference, our system demonstrated robust performance. The results yielded a MAE of 8.21 mmHg ($\pm$6.51 mmHg) for SBP and 6.66 mmHg ($\pm$4.76 mmHg) for DBP. The MAD was 7.15 mmHg for SBP and 6.25 mmHg for DBP. These findings suggest that our system can maintain reasonable accuracy and stability even when subjected to minor, real-world motion artifacts.

VI. RELATED WORK

A. Contact BP Monitoring

The volume clamp method measures blood pressure by periodically compressing the finger artery, providing continuous monitoring, but long-term use can lead to numbness or pain in the measurement site, restricted blood circulation, and increased discomfort in patients [50], [51].

Photoplethysmography (PPG), as a non-invasive technology that has achieved monetization applications, has obvious limitations, including strict reliance on direct skin contact, susceptibility to skin color, and temperature [12], [13].

As a high-precision blood pressure measurement technology, ultrasound can accurately capture arterial wall movement and blood flow velocity changes, providing continuous and detailed blood pressure dynamic information. However, ultrasound technology still requires the probe to be in close contact with the skin, and may also require the use of ultrasound coupling agents to maintain signal quality, making it difficult to popularize in daily monitoring scenarios [5].

These methods together reveal the long-standing technical paradox in the field of non-invasive monitoring - the difficulty of balancing measurement accuracy and patient comfort.

B. Radar-Based BP Methods

Recent advances in radar technology have enabled non-contact blood pressure monitoring with significant advantages over traditional approaches. These methods can be categorized into three main approaches:

1) Single-Point Measurement: mmBP [52] utilizes millimeter-wave radar with Delay-Doppler Feature Transform and adaptive filtering to capture wrist pulse displacements, extracting six pulse waveform features for blood pressure estimation. airBP [42] employs millimeter-wave radar with auto-correlation coefficient maximization beamforming to detect wrist arterial pulses. BP^3 [18] combines millimeter-wave pulse sensing with a multimodal deep learning model featuring a physiology-pulse attention mechanism applied to a large cardiovascular patient dataset. RF-BP [53] implements Ultra-Wideband radar technology with peak detection and Principal Component Analysis, using a deep learning model with multi-scale feature extractors.

2) Multi-Point Measurement: hBP-Fi [23] utilizes a single radar with super-resolution beam composition to detect pulse waveforms at multiple points along the arm, applying a blood pressure-specific transfer function. Cho et al. [26] implement dual ultra-wideband impulse radars to accurately measure pulse transit time between different arterial sites. Singh et al. [25] employ dual-radar technology to capture pulse signals from different body locations for improved BP estimation. Zheng et al. [11] develop a non-contact calibration-free approach using multi-point arterial pulse wave detection. Suzuki et al. [20] propose a non-contact system for monitoring relative blood pressure through multiple measurement points.

3) Continuous Blood Pressure Monitoring: Geng et al. [24] employ a single millimeter-wave radar, attempting to derive and disentangle distinct physiological signals from both neck and chest regions for continuous blood pressure monitoring. This approach of isolating two separate regional dynamics from a single sensor stream presents inherent challenges in signal processing and robustness, and in practice, it often necessitates breath-holding by the subject during measurement. Furthermore, the method typically relies on per-subject calibration parameters, potentially limiting its practical applicability and generalization across individuals. WaveBP [10] implements a hybrid Transformer model with beam-forming data augmentation to predict continuous arterial pressure waveforms. Despite these advancements, current radar-based approaches predominantly focus on discrete measurements while underutilizing the rich physiological information in radar signals. Most methods rely on end-to-end prediction models that neglect established hemodynamic mechanisms or require individual calibration. These limitations highlight the need for approaches that integrate multiple physiological markers to achieve reliable, calibration-free continuous monitoring of systolic and diastolic blood pressure in real-world settings.

VII. Conclusion

This study introduced Physio-BP, a dual-radar system enabling continuous, calibration-free blood pressure monitoring. By capturing pulse waves from the heart and carotid arteries, it accurately estimated systolic and diastolic pressure while mitigating motion artifacts. A novel signal selection algorithm enhanced robustness, and extensive experiments confirmed its superior accuracy over existing state-of-the-art (SOTA) methods. Physio-BP advances non-contact hemodynamic monitoring, paving the way for real-world healthcare applications.

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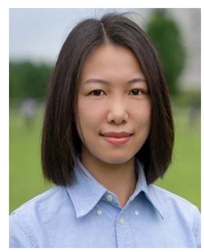

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